Lab 3: Component Modelling & Architectural Pattern Selection

Written Justification — Self-Service Coffee Kiosk System

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Architecture Selection: We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice

The system is organized into three horizontal layers: a Presentation Layer (User Interface Component) that handles touchscreen interaction, a Business Layer (Order Manager, Payment Service, and Receipt Printer Components) that implements the ordering, payment, and receipt logic, and a Data Layer (Database Component) that stores menu and pricing data. Each layer depends only on the layer directly below it through well-defined interfaces.

Two Reasons for This Choice

1. **Fixed, well-defined scope:** the kiosk supports exactly 3 coffee types, 2 sizes, and one payment method on a single embedded terminal. This bounded functionality does not need the independent scaling or distributed deployment that Microservices are built for, nor does it involve multiple remote clients that would justify a Client-Server split — a Layered design cleanly matches a single self-contained device.
2. **Clear separation of concerns for hardware integration:** the kiosk must coordinate touchscreen input, credit-card payment hardware, a receipt printer, and stored menu/pricing data. Layering isolates each concern (UI, order/payment/print logic, data) so any one of them — e.g., swapping the receipt printer driver — can change without rewriting the others, which is harder to guarantee in a tightly distributed Microservices setup for a single physical device.

Security Advantage

The Presentation Layer never talks to the Payment Service directly — it must go through the Order Manager Component's mediated Order Interface. This confines credit-card processing logic and database access entirely to the Business and Data Layers, so a compromised or buggy touchscreen UI cannot reach payment or stored pricing data directly, reducing the attack surface exposed to untrusted user input.

Performance Benefit

Because all layers run in-process on the same embedded kiosk hardware, calls between the User Interface, Order Manager, Payment Service, Receipt Printer, and Database Components are local function/interface calls rather than network requests. This avoids the network latency and serialization overhead that a Microservices architecture would introduce, keeping the order-to-receipt flow fast even on modest kiosk hardware.